

Bug in my Software

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Figure 1: Visitors interact with the installation. The XR overlays the virtual ecosystem with the physical component. ©Alina Nazmeeva

Abstract

This article discusses the conceptual underpinnings, creative process, and research methods used to create the immersive installation titled *bug in my software*. *bug in my software* reflects on the tensions between gender, labor, and technology by drawing parallels from the engineered lifecycle and genetically modified DNA of a silkmoth to the womxn-dominated textile labor industry and gendered biases embedded in AI development. The project combines textile installation, extended reality, and a screen-based speculative

ecosystem simulation. Here, the silkmoth—both organic and an engineered creature — serves as a proxy to examine the relationship between gender, labor, human and non-human intelligence.

CCS Concepts

• Applied Computing; • Arts and humanities;; • Computing methodologies; • Artificial intelligence; Modeling and Simulation;

Keywords

Installation, Textile, Artificial Intelligence, Gender

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1 ON BUGS

My software is a garden full of bugs. Bugs make it work. The bugs in the software are the products of technological development and biological manipulation, artificial intelligence, and neural renderers. They fly, roam, explore, and communicate with their environment. The garden and the bugs co-evolve, mutate, and transform. The bugs are a product of their *umwelt* and their lifeworld is the product of their lifecycle.

The term “bug” has been used in the engineering lexicon for over a century. In the 1892 *Standard Electrical Dictionary*, it was defined as a fault in an electric apparatus. In the early 1940s bugs were found in both the hardware of the first computers and within the programs that ran on them. The discovery of a moth in the Mark II computer at Harvard University in the late 1940s literalized the concept of a computer ‘bug,’ connecting the term to the domain of computing. The engineers, among whom was a computer programming pioneer Grace Hopper, taped the moth in their logbook and added a caption, “First actual case of bug being found [Hopper, 1981].” Further, the words “bug” and “debug” became a standard part of the computer programming lexicon. This origin story of the computer bug is a story of human-machine triumph - the bug is successfully removed so the program carries on.

Legacy Russel’s “*Glitch Feminism*” explores the concept of error in a machine by adopting the closely related term “glitch”. In the society occupied by machinic anxiety that fears errors and malfunctions, *glitch feminism* suggests that glitches in the already problematic system—marked by economic, racial, social, and globalized injustices—may be corrections rather than errors. A glitch, therefore, becomes a refusal to perform, a rupture in the system that manufactures and commodifies human and non-human bodies [Russel, 2020].

Yet there is a difference between a bug and a glitch. In the context of software development or videogames, a glitch marks a mystery, an exploitable possibility, or an abnormal accidental artifact, while a bug implies a mundane and annoying error, or something small and insignificant. The bugs, unlike glitches, are intended to be solved and made obsolete. Bugs disrupt the “intended” system. The buggy system can still work, yet not as planned. *Bugs transform the system they infest into something else.*

2 ON SILKMOTHS

bug in my software explores the notion of bug, agency, and human/non-human intelligence through a real-time speculative ecosystem simulation. The protagonists in my software—bugs—borrow from one of the earliest creatures to have been domesticated over 10,000 years ago: the Bombyx Mori silkworm. Unlike computer bugs, silkworms are harvested, instrumentalized, and made useful for human purposes. Throughout millennia, the silkworm has been altered by humans to the point of losing their ability to smell, see, and fly. Genes enhancing silk production have been prioritized, making them entirely dependent on humans for survival and even mating, as they are incapable of flying or functioning independently.

The industrial silk production orchestrates the life and death of these creatures. Female silkworms can lay over 300 eggs at once. These eggs hatch to form silkworms, nourished by mulberry leaves

until the pale creatures are plump and ready to spin their cocoons. Manually relocated by caretakers into a compartmentalized rack, each larva begins to metamorphose, deconstructing their bodies cell by cell to transform into a moth. Once the cocoons are fully formed, the developing moths inside are boiled or steamed alive, turning their cocoons into tombs. This ensures the precious silk can be harvested intact; otherwise, the moths would carve through the cocoon, damaging the silk thread. A few of the cocoons are preserved for the reproduction and the repetition of the cycle.

After thousands of years of selective breeding and genetic modification silkworms now resemble cute Pokémon more than insects. Yet their adorable appearance does not diminish the violent cycles of silk production. How has intentional breeding to increase silk yield also changed moths’ appearance to evoke an affective response in humans?

The connection between cuteness and violence found in the fictional world of the Pokémon franchise resonates with the practices of silkworm domestication. In Satoshi Tajiri’s world, Pokémon—pocket monsters—possess powers attuned with their ecological environments. Human Pokémon trainers and collectors exploit these traits for the purpose of battles, training and breeding Pokémon species. As Jason Bainbridge notes [Bainbridge, 2014], Pokémon franchise critiques and negotiates the relationship between environment and economy, care and dominance, non-human and human.

Similarly, silkworms embody monsters as defined in *Arts of Living on a Damaged Planet* [Tsing et al., 2017], where monsters are both the wonders of symbiosis and the threats of ecological collapse. As the editors Heather Swanson, Anna Tsing, Nils Bubandt, and Elaine Gan write, “they help us pay attention to ancient chimeric entanglements; and they point us toward the monstrosities of modern Man [Tsing et al., 2017].” The monstrosity of the silkworm is revealed in its complete shift from its early multispecies entanglements to other “non-innocent” entanglements with globalized economy and ecological disruption. Introduced by Donna Haraway, “non-innocence” acknowledges that no entity exists in a state of purity or moral clarity; all beings—humans, animals, and technologies—are entwined in webs of interdependence, responsibility, and complicity.

While sericulture—the production of silk—is regarded as critical farming practice that supports lower income populations in the countries of Global South and Asia [Ekka and Bais, 2023], it disrupts ecosystems worldwide and manufactures contained farm environments that inevitably fail under the pressure of environmental fluctuation and climate change. It relies on labor- and energy-intensive processes, from farming mulberry trees and silkworms to transforming silk cocoons into fabric. Silkworms do not survive extreme environmental fluctuations, require rigorous cleaning routines, and are severely affected by pesticides from their primary food source: mulberry leaves [Offord et al., 2016]. White Mulberry (*Morus Alba*)—indigenous to China and India—is an invasive species in North America, introduced to the continent in the 19th century by would-be sericulture industrialists. Triggering the so-called “Mulberry Craze”, a financial bubble in which the value of mulberry trees surpassed their use-value in silk production, North American sericulture failed due to a convergence of economic and environmental factors [Field et al., 2007]. Today White Mulberry continues to disrupt native ecosystems and occupy disturbed lands.

Jin Bervin's *Silk Poems* make Bombyx Mori a protagonist, telling their life story from an egg, to larva, silkworm, silkmoth, and through the *silk* DNA in fabric continue its presence around and under human skin [Bervin, 2019]. The themes of life and death and "non-innocent" entanglements between humans, Bombyx Mori, and White Mulberry, among myriads of other species, permeate the work. Unlike *Silk Poems*, *bug in my software* works with the *silkmoth* DNA, imagining the silkworm's lifecycle post-human interference, in other kinds of "non-innocent" entanglements not predicated on globalized economies and environmental degradation. The work asks, what happens when the engineered beings are left alone?

3 ON SOFTWARE

Silkmoths and their lifecycle embody the intersection of biological manipulation, technological advancement, extractive power, and extracted labor. This example of industrialization of natural processes parallels the exploitation of gendered biases whether in textile production or AI development.

Textile work—weaving, sewing, embroidery, quilting, or knitting—is often undervalued, considered either as unpaid domestic labor or low-paid, womxn-dominated vocational work. In the art world, textile works are seen as "low art", compared to the "high art" of painting and other male-dominated mediums. Professional textile work is recognized when performed by men but is deemed "amateur" if done at home by women, even if compensated [Parker, 1984].

Textile, especially weaving—a domestic and indigenous craft—has been acknowledged for its deep connection with computation [Fernaes et al., 2012], while womxn and other minoritized people are often not recognized for their contribution to the field of computation. The Jacquard loom, with its punch cards used to create woven patterns, is a precursor to modern electronic computational devices. These punch cards were later used by Charles Babbage, with Ada Lovelace, the world's first computer programmer, in their work on difference and analytical engines. Their development laid the path for the first programmable computer, marking a clear throughline from weaving to computation.

Womxn have played crucial but often unacknowledged roles in the early computation, frequently taking on jobs deemed tedious and unprestigious by men, such as computers, circuit makers, and programmers [Evans, 2018]. As these roles gained prestige and higher pay, men often took them over, turning them into highly competitive, sought-after positions.

This dynamic is mirrored in the role of gender in the development of Artificial Intelligence. Voice assistants, caregiving bots, and virtual helpers, often feminized in voice, form, and function, are designed to be docile, soothing, accommodating, and perpetually available. Many robots are designed to function as maids, personal assistants, or care workers. Professor Safiya Noble, the author of *Algorithms of Oppression*, argues that increasing interaction with female virtual assistants diminishes women to "a gendered female who responds on-demand" [Robison and Chin-Rothmann, 2020].

The design of virtual assistants is grounded in patriarchal, commodified, manufactured and symbolic femininity. Most leading voice assistants are feminized or female by default, both in name

and in sound of voice. Apple's Siri, named from a Norwegian word meaning 'beautiful woman who leads you to victory,' and Amazon's Alexa, named after the Library of Alexandria, both have female voices by default. Microsoft's Cortana is named after a fictional female AI from the video game Halo. Even Google Assistant, though gender-neutral in name, uses a female voice.

Companies often justify this by citing academic research that shows that people prefer a female to a male voice. However, Siri defaults to a male voice in certain languages, such as Arabic, British English, Dutch or French suggesting deeper gendering at play. Perhaps there is more intention for specific gendering beyond a trope of preference of universal preference of female voice. Following the critique, Apple removed Siri's default gender in 2021, allowing users to choose during setup.

In *The Institute for Other Intelligences*, Machinka Firunts Hakopian envisions AI through Artificial Killjoys—machines that challenge biases and omissions in their training, addressing gendered AI and the labor behind it. Inspired by Sara Ahmed's concept of the feminist killjoy, these machines generate "multiple ways of knowing" and "disrupt what was previously known [Hakopian, 2022]." The bugs in *my software* act as killjoys too, by reminding the visitors of the tensions between gender, labor and technology.

4 BUGS IN MY SOFTWARE

bug in my software pairs a physical textile installation and a real-time speculative ecosystem simulation full of bugs: the nuisances, the monsters, the killjoys, the crucial actors of the ecosystems. The physical installation employs the silkmoth's DNA sequence, transformed into a visual pattern that is both a blueprint and a historical document of the anthropogenic transformation of the species. This document is expanded into an architectural scale landscape intended to be occupied, inhabited, and lounged on.

The Bombyx Mori DNA pattern was visualized as a color field, with each chemical base color-coded and further transformed using generative AI. The resulting pattern was extended with soft wool yarn structures that give form to the genetic imprints altered by human hands. While automation technology drove the creation of the pattern, the woolen structures were handcrafted by the artist over several months. This process highlights the tension between human labor and automation, in which the computational labor (and the hidden human labor that grounds it) is made invisible. Manual, repetitive, and prolonged labor of textile production embodies the hidden human labor that enables AI and automation and underscores the selective human-made evolution of the silkmoth.

Imitating the logic of three-dimensional simulation, where the surfaces are infinitely thin and single-sided, the mirrored plexiglass edges of the physical installation create an optical illusion of a floating virtual surface that does not have any substrate. Like the interplay between labor and automation, here the work gestures towards the tension between the seemingly infinite digital landscape and the finite matter that powers it.

While immersed in the soft limbs of handmade textiles, visitors are invited to observe the speculative ecosystem unfolding atop the textile landscape, revealed through XR headsets and multiple screens scattered throughout the installation. The bright colors of the DNA-patterned textile echo the aesthetics of the simulation,



Figure 2: *bug in my software* installation full view. (©Alina Nazmeeva)

exploring the tension between the organic and the engineered—between “nature” and “artifice.” The blocky, fuzzy, hallucinatory aesthetics of the simulation gestures towards the limitations—*bugs*—and incompleteness of digital spaces, echoing the cyberfeminist critique of blind spots and failures of mainstream technological advancements [Russell, 2020]. Rooted in the discussion of agency and authorial control, the aesthetics are both shaped by the artist’s choices and influenced by AI-generated textures. The soundscape is composed with cricket-like electronic glitch sounds, emitted as the bugs transition from eggs to moths. Rejecting concrete, defined, fixed, photo-realistic visual language, the work contrasts the embodied experience of the physical textile landscape with the low-fidelity, staggered motions of the buggy speculative ecosystem. *bug in my software* converges textile labor with experimental AI workflows and AI-agent driven ecosystem simulation, exploring collaboration with AI and its connection to the non-human world and ecosystemic relationships. Among the AI agent-driven virtual ecosystem evolving atop of the physical landscape, the bugs are the protagonists of this world and are the products of the confluence of technological and ecological imaginations, word embedding systems, text-to-image, text-to-3d, and image-to-3d AI models used in different combinations.

In *Insect Media* Jussi Parikka outlines the confluence between insects and computation, suggesting that technology, in particular digital technology and media, is inherently nonhuman [Parikka, 2010]. The cross-pollination with ethology, and even entomology

- the science of bugs - informed a non-human understanding of technology, in which the swarm behavior and distributed intelligence attributed to insects is considered as the optimal model for computational problem-solving. Object-oriented, emergent computation, boid algorithms, and AI, among others suggest modes of communication, language, and becoming, beyond human capacity and cognition. The bugs in *my software* apply some of these computational frameworks to emerge, grow, live, coalesce, interact, and evolve, die, and decompose in the speculative lifecycle. They are equipped with artificial vision that directs them to trees, food sources, and other creatures, and are entangled in the process of co-becoming with their environment.

In this garden, human visitors are mere observers, sometimes followed by curious creatures, creating ripples in the landscape. Here, the AI agents are weavers of a new world, spinning the detritus of waste into the web of an emergent ecosystem. Like their biological counterparts, the bugs are locked in “non-innocent” entanglements of global economy; as moths in the simulation die, they leave behind piles of electronic waste.

There is no specified run time for the simulation. When the real-time simulation ultimately collapses—whether due to hardware limitations, the destruction of the bug population, or an excess of e-waste—it resets and perpetually runs, failing again and again. In *bug in my software*, text messages appear each time the simulation restarts. These fleeting messages are co-created by the author and an AI, using a custom-trained language model with a vocabulary

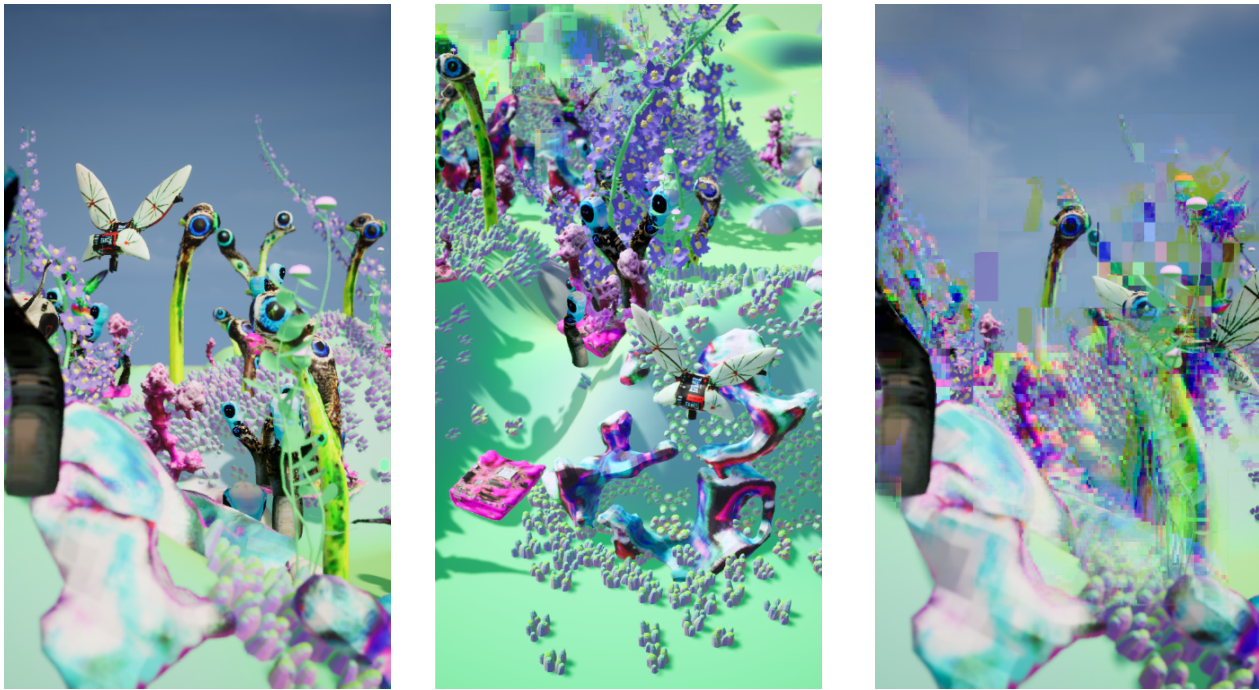


Figure 3: Screenscaptures of the simulation. (©Alina Nazmeeva)



Figure 4: The screens and XR headset display real-time simulation. (©Alina Nazmeeva)

drawn from the project's core concepts. After a few seconds, the messages vanish, and the simulation begins anew.

5 BUG IN MY SOFTWARE: EPILOGUE

This cycle of endless restart and the impossibility of total collapse echoes the concept of *Permalife* offered by Bo Ruberg. In videogames and interactive media, permalife makes it impossible



Figure 5: The textile pattern represents a visualization of Silkmoth DNA. The wool structures shape the genetic imprints, creating a tactile landscape for visitors to inhabit. (©Alina Nazmeeva)

for players to die. Found in works by independent LGBTQ+ game creators, permalife games explore queer experiences, forcing players to exist, survive and persevere in collapsing, challenging, and overwhelming worlds. As Ruberg notes, permalife can be seen as a refusal to death, not as an optimistic escape, but as a way of enduring within a messy, mournful, and complex reality [Ruberg, 2017].

Permalife games resonate with the position of Donna Haraway's book *Staying with the Trouble*. Haraway suggests that survival in the Chthulucene requires embracing complexity, staying present with challenging issues, and fostering symbiotic relationships with other species [Haraway, 2016]. Chthulucene is introduced by Haraway as a critical term that decenters the human-centered perspectives of Anthropocene and Capitalocene. This involves recognizing the agency of nonhuman entities.

The real-time simulation of *bug in my software* delves into the tension between control and autonomy in the interplay between human and non-human intelligences and beings. In this simulated garden, silkmoths regain their ability to fly and roam, grow, and communicate. These AI creatures develop emergent behaviors based on their needs to socialize, explore, and feed, leading to unpredictable actions beyond the artist's control.

This raises a question; how can we understand autonomy and agency of the digital beings in the simulation context? While the silkmoths appear to have autonomy, their behavior is still governed by parameters set by the artist, including initial conditions of the AI environment. The simulation's setup determines how the garden begins, what plants are present, the number of bugs, and their interactions. Thus, while the creatures exhibit emergent behaviors, their

agency exists within the constraints of the system's configuration and the simulation state.

Combining the physical textile landscape with real-time simulation, *bug in my software* confronts the conventional ways humans interact with emergent technologies, their promises, and their inherent prejudices. Recognizing that gender, technology and the notion of intelligence are co-constructed, and following the concept of permalife, the visitors are invited to embed (or re-embed) themselves into the garden: find a position, feel the ground, envelop in the landscape, contemplate, cuddle, and rest, while the creatures work, threads weave, and GPU fans spin.

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References

- Grace Murray Hopper. 1981. The First Bug. *Annals of the History of Computing*, vol. 3 #3, 1981, pp. 285–286.
- Legacy Russell. 2020. *Glitch Feminism: A Manifesto*. London: Verso.
- Jason Bainbridge. 2014. 'It Is a Pokémon World': The *Pokémon* Franchise and the Environment. *International Journal of Cultural Studies* 17, no. 4: 399–414. <https://doi.org/10.1177/1367877913501240>.
- Anna Tsing, Heather Swanson, Elaine Gan, and Nils Bubandt, eds. 2017. *Arts of Living on a Damaged Planet: Ghosts and Monsters of the Anthropocene*. University of Minnesota Press.
- Vinod Ekka, and Pratima Bais. 2023. Contribution of Sericulture in the Socio-Economic Transformation of the Workers. *Asian Journal of Applied Science and Technology* 07 : 217–24. <https://doi.org/10.38177/ajast.2023.7223>.
- C. Offord, Fritz Vollrath, and Chris Holland. 2016. Environmental effects on the construction and physical properties of *Bombyx mori* cocoons. *J Mater Sci* 51,

- 10863–10872. <https://doi.org/10.1007/s10853-016-0298-5>
- Jacqueline Field, Marjorie Senechal, and Madelyn Shaw. 2007. *American Silk, 1830-1930: Entrepreneurs and Artifacts*. Texas Tech University Press.
- Jen Bervin. 2019. *Silk Poems*. Hong Kong University Press.
- Rozsika Parker. 1984. *The Subversive Stitch: Embroidery and the Making of the Feminine*. London: Women's Press.
- Ylva Fernaeus, Martin Jonsson, and Jakob Tholander. 2012. Revisiting the Jacquard Loom: Threads of History and Current Patterns in HCI. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 1593–1602. Austin Texas USA: ACM. <https://doi.org/10.1145/2207676.2208280>.
- Claire Lisa Evans. 2018. *Broad Band: The Untold Story of the Women Who Made the Internet*. New York, New York: Portfolio/Penguin.
- Mishaela Robison and Caitlin Chin-Rothmann. 2020. How AI Bots and Voice Assistants Reinforce Gender Bias. *Brookings*. <https://www.brookings.edu/articles/how-ai-bots-and-voice-assistants-reinforce-gender-bias/>.
- Mashinka Firunts Hakopian. 2022. *The Institute for Other Intelligences*. X Topics. South Pasadena, CA: X Artists' Books.
- Jussi Parikka. 2010. *Insect Media: An Archaeology of Animals and Technology*. Posthumanities. Minneapolis: University of Minnesota Press.
- Bonnie Ruberg. 2017. Permalife: Video Games and the Queerness of Living. *Journal of Gaming & Virtual Worlds* 9, no. 2: 159–73. https://doi.org/10.1386/jgvw.9.2.159_1.
- Donna Jeanne Haraway. 2016. *Staying with the Trouble: Making Kin in the Chthulucene*. Experimental Futures. Durham: Duke University Press.